Code No: 117DU **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, April/May - 2018
IMAGE PROCESSING AND PATTERN RECOGNITION
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Define Histogram [2]
b) Differentiate linear spatial filter and nonlinear spatial filter [3]
c) List out the properties of the second derivative around an edge [2]
d) What is data redundancy? Explain three basic data redundancy? [3]
e) What is the need for compression? [2]
f) What are the operations performed by error free compression? [3]
g) Define shape numbers [2]
h) Define compactness. [3]
i) What is pattern class? [2]
j) What are the applications of pattern recognition? [3]

PART-B

(50 Marks)

2.a) Explain the principle of sampling and quantization. Discuss the effect of increasing the i) Sampling frequency ii) Quantization levels on image
b) Explain image enhancement in the frequency domain. [5+5]

OR

3.a) Explain the various sharpening filters used in spatial domain.
b) Explain Homomorphic filtering in detail. [5+5]

4.a) Discuss in detail about Hit or Miss Transform
b) Explain the operations i) Opening and Closing ii) Dilation and Erosion. [5+5]

OR

5.a) Explain the concept of thresholding in image segmentation and write its merits and demerits.
b) Briefly discuss about the segmentation techniques that are based on finding the regions directly. [5+5]

JNTUH USED 08-05-18 AM Set 2

6.a) Discuss in detail about digital image watermarking
b) Explain about Lossy compression. [5+5]

OR

7.a) Differentiate between lossless and lossy compression and explain transform coding system with a neat diagram
b) Explain Lossless predictive coding. [5+5]

8.a) What are the principal components of description? Explain.
b) Explain the regional descriptors in image representation. [5+5]

OR

9.a) What are chain codes? Discuss the demerits of chain codes
b) State the various polygonal approximation methods. [5+5]

10.a) With a neat diagram explain a simple automatic pattern recognition method.
b) Explain the pattern and pattern classes in object recognition. [5+5]

OR

11.a) With necessary equations explain about K-Means clustering.
b) Discuss the fundamental problems encountered in pattern recognition system. [5+5]

--ooOoo--

JNTUH USED 08-05-18AM Set 2